Matthew Amiadamen

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Education

Dublin City University May 2027
BSc in Computer Science *Glasnevin, Dublin 11*
- Relevant Courses: Advanced Algorithms, AI, Data Mining, Full Stack Development, Linear Algebra, Statistics

Experience

consulting.ie – Software Engineer (Part-Time) September 2026 – Present
Part-time alongside final year of BSc Computer Science *Dublin*
- Worked directly with CTO to architect **Pulse**, a multi-tenant enterprise agentic ERP (TypeScript, Fastify, PostgreSQL), co-authoring 20+ architecture decisions on auth, embeddings, vector search and row-level security.
- Designed LLM routing across **Amazon Bedrock** (Claude models) and a **Cerebras** fast tier for sub-1.5 s voice and triage turns, with per-tenant cost metering and prompt caching projected to cut repeated-prompt input cost by ∼60%.
- Implemented tenant-scoped B-tree indexing (70+ composite indexes) so multi-tenant lookups run as logarithmic binary-search seeks instead of full scans, alongside vector search over Bedrock embeddings.
- Built client-app integrations through a **Composio** bridge (39-toolkit catalogue): allowlisted, version-pinned actions, idempotent writes, per-tenant rate-limit sharing, and OAuth tokens held in AWS Secrets Manager.
- Built data integration for a 70-staff, 19-coach operation with 500k+ bookings: a PostgreSQL mirror at 2–4 s replication lag, live fleet telematics ingest, and 8,000+ documents synced into a RAG index.
- Cut assistant latency by up to 44% (14.9 s to 8.4 s) on a production agentic ERP (ECS, S3, EventBridge) by routing lookups to a fast-model lane and parallelising retrieval with intent classification (159-question eval suite).

Dublin City University – UN Sustainable Development April 2025 – September 2025
Undergraduate Research Assistant, Prof. Hyowon Lee *Dublin*
- Trained NLP complaint-triage classifiers (Python, spaCy, scikit-learn) for Cabhar, a UN SDG 8 platform for reporting workplace exploitation, routing reports to NGOs at 98% accuracy across 11,000 complaints per year.

Projects

GrianView – Belfast Solar Intelligence Platform
Technologies: Next.js 14, React, Three.js, Mapbox GL, ArcGIS, OpenAI API, Overpass API
- Worked with a PhD researcher at Queen's University Belfast on a free solar-suitability service for any Belfast home, running ArcGIS solar radiation over the city's LiDAR surface model to profile 52,872 buildings.
- Built a 5-factor weighted scorer, an in-memory index for instant search over 52k rows, live 3D Three.js rendering from OpenStreetMap footprints, and LLM-written payback and CO_2 reports.

Lámha – Craicathon Hackathon – *Create It Track Winner*
Technologies: Python, TensorFlow, OpenCV, MediaPipe Hands, KNN Classifier
- Built a winning Irish Sign Language tutor now used in Dublin post-primary schools, hitting 85% sign classification accuracy with real-time in-browser gesture recognition (MediaPipe landmarks, custom KNN) and no paid APIs.

Mayfain – Verified-Identity Layer for AI Agents – *Pressure Cooker, Dublin AI Week 2026*
- Developed institution-side MCP servers and a personal-data vault an AI agent completes applications from without the model seeing raw data; selected as one of 10 teams.

Competitions & Awards

Bloomberg bpuzzled – 1st Place *April 2025*
- Part of team Cubits, scoring 8/9: at the time, the highest score by any Irish university team in competition history.

Extracurriculars & Leadership

DCU Debate Society: Events Officer & Committee Member • Irish Times National Debating Competition
Sport: DCU Rugby Club Forwards Captain • Bohemians Limerick RFC
Other: UL Cyber Security Camp • Tidy Towns Limerick • Waiter • Bartender • Phonewatch Sales Rep

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL
Cloud & Infra: AWS (Bedrock, ECS Fargate, S3, EventBridge, RDS, SQS), Docker, GitHub Actions CI/CD
Frameworks & Data: Node.js, Fastify, Next.js, React, Django, Supabase, PostgreSQL (row-level security), MongoDB
AI/ML & Testing: LLM routing, RAG, embeddings, vector search, evals, TensorFlow, scikit-learn, mutation testing